DR1 Calculations — Portable Weather Station

Rokas Ulys · VIKO EIF · Course: Electronic Control Systems · Revision 2026-11-09

Board A: control node (STM32F105R8T6). Board B: sensor node (STM32G474RET6). Link: SX1262 868 MHz radio + CAN. Designators refer to the schematics after the final annotation (2026-11-09).

HSE crystals (8 MHz)

Board A Y2 and board B Y3 (Fox FOXSLF/080-20, 8 MHz, HC-49S, CL = 20 pF) work with 33 pF load capacitors: effective load ≈ 19.5–21.5 pF, gain margin ≈ 42 (required > 5).

Parameter	Value	Source
Crystal load capacitance CL	20 pF	FOXSLF/080-20 datasheet
Stray capacitance Cs (MCU pins + PCB)	3–5 pF	ST AN2867, sec. 3.3 (assumption)
Crystal ESR	80 Ω	assumption, check Fox datasheet (TBR)
Crystal C0	7 pF	assumption, check Fox datasheet (TBR)
MCU gm (HSE)	STM32F105: 25 mA/V; STM32G474: Gm_crit_max 1.5 mA/V	DocID15274 Table 22; DS12288 Table 41
Load capacitors CL1 = CL2	33 pF, C0G	schematic: A C12/C13, B C23/C24

1. Load capacitors (AN2867, sec. 3.3):

$$C_{L1} = C_{L2} = 2\,(C_L - C_s) = 2\,(20 - 5) = 30\ \mathrm{pF} \quad (C_s = 3\ \mathrm{pF} \Rightarrow 34\ \mathrm{pF})$$

The nearest E12 value, 33 pF, was selected. Actual load: 33/2 + Cs = 19.5–21.5 pF, i.e. ±1.5 pF from 20 pF.

2. Gain margin (AN2867, sec. 3.4):

$$g_{m,crit} = 4 \cdot ESR \cdot (2\pi f)^2 \cdot (C_0 + C_L)^2 = 4 \cdot 80 \cdot (2\pi \cdot 8\,\mathrm{MHz})^2 \cdot (27\,\mathrm{pF})^2 \approx 0.59\ \mathrm{mA/V}$$

Board	MCU parameter	Check	Result
A (STM32F105)	gm = 25 mA/V	gm / gm,crit = 25 / 0.59 ≈ 42 > 5	OK
B (STM32G474)	Gm_crit_max = 1.5 mA/V	gm,crit = 0.59 mA/V < 1.5 mA/V	OK

The G474 datasheet gives the maximum allowed crystal gm,crit rather than the oscillator gm, so per AN2867 the calculated value is checked against it. ESR and C0 will be taken from the Fox datasheet and recalculated (TBR).

Data flow (group 4 requirement)

Board B produces 52 B/s (one record per second), while the radio in the EU 868.0–868.6 MHz band with a 1 % duty cycle can only send ~350 packets per hour. Therefore 1-minute summaries are sent over the radio (0.17 % duty cycle), all raw data is stored in board B's 8 MB flash (~44 h) and offloaded over CAN.

1. Source — size of one record (1 Hz)

Field	Sensor	Bytes
Timestamp	RTC (LSE + RTC_VBAT)	4
Temperature, humidity	SHT40	4
Pressure, temperature, humidity	BME280	8
Illuminance	VEML7700	4
CO2, temperature, humidity	SCD41 (updates every 5 s)	6
eCO2, TVOC	CCS811	4
Differential pressure (wind) and computed speed	SDP810-125Pa	4
Acceleration X/Y/Z, gyroscope X/Y/Z	LSM6DSOX	12
Hall sensor state and pulse count	SM351LT	2
Battery voltage (ADC)	STM32G474	2
CRC	—	2
Total		**52**

Data rate: 52 B/s = 416 bit/s ≈ 187 kB/h ≈ 4.5 MB/day.

2. CAN capacity (A ↔ B, 500 kbit/s)

A standard CAN 2.0 frame with 8 data bytes takes ~111 bits, worst case with bit stuffing ~135 bits. Useful throughput ≈ 500 000 / 135 × 8 B ≈ 29.6 kB/s, i.e. ~570 times the source rate. CAN is not the bottleneck: the full 8 MB memory is offloaded in ~4.5 min.

3. Radio capacity (SX1262, SF7, BW 125 kHz, CR 4/5)

Symbol time and number of payload symbols (Semtech SX1261/2 datasheet, sec. 6.1.4):

$$T_{sym} = \frac{2^{SF}}{BW} = \frac{128}{125\,\text{kHz}} = 1.024\ \text{ms}$$

$$n_{payload} = 8 + \left\lceil \frac{8 \cdot 52 - 4 \cdot 7 + 28 + 16}{4 \cdot 7} \right\rceil \cdot 5 = 8 + 16 \cdot 5 = 88$$

$$T_{air} = (8 + 4.25 + 88) \cdot 1.024\ \text{ms} \approx 103\ \text{ms}$$

EN 300 220-2 sub-band M (868.0–868.6 MHz): 25 mW e.r.p., duty cycle ≤ 1 % = 36 s per hour. Maximum packets: 36 s / 0.103 s ≈ 350 packets/h, i.e. one packet every ~10 s. Sending every 1 s record would need ~10 % duty cycle — 10 times too much.

4. How the gap is closed

Path	Throughput	Solution
Source → B memory	52 B/s	all raw records go to an IS25LP064A ring buffer
B → A by radio	1 summary (avg/min/max) per 1 min, 52 B	103 ms/min = 6.2 s/h = 0.17 % duty cycle
Margin for link recovery	36 − 6.2 ≈ 29.8 s/h ≈ 290 packets/h	after a link outage A receives the buffered summaries
B → A over CAN (cable)	29.6 kB/s	raw data offload during service
A → SD card	52 B/min	far below SD SPI write speed

5. B memory capacity (outage duration)

$$t_{buf} = \frac{8 \cdot 2^{20}\ \text{B}}{52\ \text{B/s}} \approx 161\,300\ \text{s} \approx 44.8\ \text{h}$$

The 8 MB memory holds ~44 h of raw data, so B can run almost 2 days without a link to A without losing a single 1 Hz record. Radio summaries (75 kB per day) would only overflow after a ~110-day outage.

Assumptions (TBR): 1 Hz logging, 52 B record, SF7/BW125 — if a higher SF is chosen (longer range), the packet becomes longer and the summary interval must be recalculated.

Power budget

Worst-case 3.3 V current: board A ≈ 390 mA, board B ≈ 426 mA. Both TPS63021 converters can supply this, and the battery current (A ≈ 0.47 A, B ≈ 0.75 A) is below the BQ29700 overcurrent limit (≈ 1.1–1.8 A). Calculated from maximum datasheet values (course topic 1, sec. 2.1).

Board A, VCC_3.3V_A (U7 TPS63021)

Load	I max	Conditions	Source
STM32F105R8	68 mA	72 MHz, all peripherals, 85 °C	DocID15274 Rev 10, Table 13
SX1262 transmit	118 mA	+22 dBm matching, VBAT 3.3 V	DS.SX1261-2 Rev 2.2, Table 3-6
microSD write	100 mA	SD default speed	SD Physical Layer Spec.
Olimex MOD-LCD2.8RTP display	~80 mA	backlight on	estimate, TBR (measure)
SN65HVD230 (CAN)	17 mA	dominant/recessive, no load	TI SLOS346O
2 LEDs (power, status)	2.6 mA	(3.3 − 2.0 V) / 1 kΩ × 2	calculation
UEXT I2C pull-ups 2 × 4.7 kΩ	1.4 mA	both lines low	calculation
Total	**≈ 388 mA**		

Board B, VCC_3.3V (U11 TPS63021)

Load	I max	Conditions	Source
STM32G474RE	~45 mA	170 MHz; 173 µA/MHz × 170 = 29 mA + peripherals	ST STM32G4 overview; TBR (CubeMX calculator)
SX1262 transmit	118 mA	+22 dBm matching	DS.SX1261-2, Table 3-6
SCD41 (peak)	175 mA	measurement pulse, 3.3 V	Sensirion SCD41 (average 15 mA)
CCS811	30 mA	mode 1, avg. max	ams CCS811 DS v1-06
IS25LP064A write/erase	25 mA	programming	ISSI datasheet, TBR
SN65HVD230 (CAN)	17 mA	no load	TI SLOS346O
SDP810-125Pa	5.5 mA	measuring	Sensirion SDP8xx
BME280, SHT40, VEML7700, LSM6DSOX	2.3 mA	measuring (≈ 1 + 0.5 + 0.05 + 0.55)	manufacturer datasheets
2 LEDs + I2C pull-ups 4 × 2.2 kΩ	8.6 mA	2.6 + 6.0 mA (all lines low)	calculation
Total (all peaks together)	**≈ 426 mA**		

Board B, VCC_5V (TPS61023) — fan

Sunon MF20100V1-1000U-A99: 5 V, 122 mA max (0.61 W). From the battery (3.0 V, η ≈ 90 %): I ≈ 0.61 / 0.9 / 3.0 ≈ 0.23 A. TPS61023 switch current limit 3.7 A — large margin.

Battery current (worst case, VBAT = 3.0 V, η ≈ 90 %)

$$I_{BAT} = \frac{3.3\ \mathrm{V} \cdot I_{3V3}/\eta + P_{5V}/\eta}{V_{BAT}}$$

Board	3.3 V power	5 V power	I_BAT	BQ29700 OCD limit
A	1.28 W	—	≈ 0.47 A	≈ 1.1–1.8 A
B	1.40 W	0.61 W	≈ 0.75 A	≈ 1.1–1.8 A

OCD limit = 100 mV / (2 × R_DS(on) AO3400A) = 100 mV / 55–90 mΩ. All peaks at the same time (SCD41 175 mA pulse + SX1262 transmit + flash write) do not coincide in practice, so the average current is much lower. Display and G474 values are TBR: to be confirmed by measurement or the STM32CubeMX Power Consumption Calculator.

Power-loss protection (board A)

An SD card write is never cut off: when the switch is turned off, the MCU keeps the power on through the BAT54C until it finishes the write, and logging stops when the battery drops to 3.3 V — 0.5 V above the BQ29700 cut-off (2.8 V).

1. PWR_SENSE_A divider (100 kΩ / 200 kΩ + 100 nF)

$$V_{PC2} = V_{BAT} \cdot \frac{200\ \text{k}\Omega}{100\ \text{k}\Omega + 200\ \text{k}\Omega} = \frac{2}{3} V_{BAT}$$

V_BAT	V_PC2	ADC code (12 bit, VREF = 3.3 V)	State
4.35 V (OVP)	2.90 V	3599	full, below the 3.3 V ADC limit
3.30 V	2.20 V	2730	logging stopped (analog watchdog)
2.80 V (UVP)	1.87 V	2317	BQ29700 disconnects
switch OFF	0 V	0	MCU finishes the write and releases PWR_HOLD_A

Resolution: 3.3 V / 4096 × 3/2 ≈ 1.2 mV of battery voltage. Divider current when ON: 4.35 V / 300 kΩ ≈ 14.5 µA; when OFF: 0 (the divider is powered only through the switch).

2. ADC sampling and time constant

Source impedance 100 kΩ ‖ 200 kΩ = 66.7 kΩ, so the ADC sampling capacitor is charged from the 100 nF capacitor. Time constant τ = 66.7 kΩ · 100 nF ≈ 6.7 ms: switch-off is detected within < 20 ms (3τ), while the battery voltage changes much more slowly.

3. EN levels through BAT54C

Source	Voltage at EN pin	TPS63021 VIH	Result
PWR_SW_A (switch ON, V_BAT ≥ 3.0 V)	≥ 3.0 − 0.3 = 2.7 V	1.2 V	enabled
PWR_HOLD_A (MCU, 3.3 V)	3.3 − 0.3 = 3.0 V	1.2 V	enabled
both low	0 V through 1 MΩ	VIL 0.4 V	disabled

The SD write busy time can reach hundreds of ms, but power is held for as long as the firmware needs, so no hold-up capacitance has to be calculated. Only battery removal during operation is not protected (noted in the documentation).

RF matching (SX1262, 868 MHz)

The discrete network values are taken from the Semtech e449 (868 MHz) reference design and checked with the AN5457 formulas: the calculated values match the reference, and the trap resonates at 1752 MHz ≈ 2 · 868 MHz.

1. Band, power, duty cycle (EN 300 220-2 V3.3.1, Table 4)

Parameter	Value
Sub-band	M: 868.0–868.6 MHz
Maximum power	25 mW e.r.p. = +14 dBm
Conducted power (antenna +2 dBd)	≈ +12 dBm
Duty cycle	≤ 1 % (36 s/h); used 0.17 % (see Data flow)
Harmonic limit 1–6 GHz	≤ −30 dBm

2. PA matching L1/C1 (AN5457, Example 7)

Zopt = 15.27 + j1.27 Ω (14 dBm), so Rout = 15.27 Ω, Xc = 1.27 Ω.

$$m = \sqrt{\frac{50}{R_{out}} - 1} = 1.508; \quad L_1 = \frac{50m/(m^2+1) + X_c}{2\pi f} = 4.45\ \text{nH}; \quad C_1 = \frac{1}{2\pi f \cdot 50/m} = 5.53\ \text{pF}$$

Check: Rout · (1 + m²) = 15.27 · 3.27 = 50 Ω. In the e449 circuit the same stage is implemented as a trap + C 5.6 pF (course topic 6, part 2, slide 14).

3. H2 trap (L 2.5 nH ‖ C 3.3 pF)

$$f_{res} = \frac{1}{2\pi\sqrt{2.5\ \text{nH} \cdot 3.3\ \text{pF}}} \approx 1752\ \text{MHz} \approx 2 \cdot 868\ \text{MHz}$$

Inductor tolerance ±0.1 nH (B grade): with ±0.3 nH the trap could shift by ~6 % and H2 would no longer be suppressed (course topic 6, slide 29).

4. π filter after the switch (C 3.3 pF – L 9.1 nH – C 3.3 pF)

$$L = \frac{50}{2\pi f} = 9.17\ \text{nH}; \quad C' = C'' = \frac{0.95}{50 \cdot 2\pi f} = 3.48\ \text{pF}$$

Nearest catalogue values 9.1 nH and 3.3 pF (e449 / e499).

5. RX balun (C 2.4 pF, L 15 nH, C 1.8 pF) — e449 reference values; topology per AN5457 sec. 5.2 (series C11, L between RFI_P and RFI_N, C12 to GND). Tuned by measurement.

6. Other RF design justifications

Decision	Justification
47 nH choke (VR_PA → RFO)	e449 / AN5457, Idc ≥ 120 mA (+22 dBm: 118 mA)
32 MHz crystal, not TCXO	packet 103 ms < 0.4 s (course topic 6, slide 11); PCB: thermal isolation
TX current +14 dBm (with +22 dBm matching)	90 mA (DS.SX1261-2 Table 3-6)
Antenna π footprint (0 Ω + 2 NP)	tuning with NanoVNA, S11 ≤ −10 dB
ESD Murata LXES15AAA1-133	0.05 pF, 15 V ≥ +22 dBm amplitude ≈ 4 V

I2C pull-up resistors

Board B 2.2 kΩ and board A (UEXT) 4.7 kΩ fall within the allowed range for Fast mode (400 kHz). Calculated per TI SLVA689 and NXP UM10204 (course topic 5, I²C design).

$$R_{p,min} = \frac{V_{CC} - V_{OL}}{I_{OL}} = \frac{3.3 - 0.4}{3\ \text{mA}} \approx 967\ \Omega; \qquad R_{p,max} = \frac{t_r}{0.8473 \cdot C_b}$$

Bus	C_b estimate	R_p,max Fast (300 ns)	R_p,max Standard (1000 ns)	Fitted	Result
B I2C1 / I2C2 (worst case: MCU + 6 sensors)	7 × 10 pF + 15 cm × 0.8 pF/cm ≈ 82 pF	4.3 kΩ	14.4 kΩ	2.2 kΩ	OK
A UEXT I2C (MCU + display + cable)	2 × 10 pF + cable ~50 pF ≈ 70 pF	5.1 kΩ	16.9 kΩ	4.7 kΩ	OK (TBR: does the module have its own pull-ups)

Current with the line low: 3.3 V / 2.2 kΩ = 1.5 mA (< 3 mA I_OL). If the Olimex module has its own pull-ups they reduce R_p in parallel — then the board A 4.7 kΩ resistors are left unpopulated (DNP). STM32 internal pull-ups (25–55 kΩ) are not suitable: in Standard mode they would allow only ~30 pF.

SPI timing budget for the SD card (board A)

The SD card runs on SPI1 at 18 MHz: the read path fits into half a period with ~9 ns margin. There are no level shifters (MCU and card both 3.3 V), so no extra delay (course topic 5, STM32G474E-EVAL microSD review).

Parameter	Value	Source
SPI1 clock	72 MHz / 4 = 18 MHz	STM32F105 SPI max 18 MHz (DocID15274)
SD default speed limit	25 MHz	SD Physical Layer Spec., ch. 2
Initialization clock	72 MHz / 256 ≈ 281 kHz (≤ 400 kHz)	SD Physical Layer Spec.
Half period	1 / (2 · 18 MHz) = 27.8 ns	calculation
Card output delay	~14 ns	TBR (full SD spec / card data)
MCU MISO setup time t_su(MI)	~5 ns	DocID15274 SPI characteristics, TBR
Traces (~3 cm)	~0.2 ns	~6.7 ps/mm FR-4
Margin	**27.8 − 14 − 5 − 0.2 ≈ 8.6 ns**	OK

The CS pull-up (R31 20 kΩ to SD_VDD_A) keeps the card deselected after reset; its RC with ~20 pF ≈ 0.4 µs — CS speed is not critical. If the margin were insufficient, SPI1 would be reduced to 9 MHz (72 / 8).

ERC messages and explanations

Both projects (A_Board_Weather.Station and Weather_Data, board B) pass Validate with 0 errors (final check 2026-11-09, after annotation). The remaining warnings are informational; every connection was checked (Alt+click and file analysis).

Board A

Message	Explanation
CAN_RX_A, RF_MISO_A: IO Pin + Output Pin	SN65HVD230 R and SX1262 MISO outputs drive MCU GPIOs configured as inputs. Normal.
NetR3_2 (BOOT0) has no driving source	BOOT0 is set by R3 10 kΩ to GND; no active driver needed.
NetR17_1, NetR22_2 have no driving source	U6/U8 configuration inputs (TS, PROG) are set by resistors.
PWR_EN_A has no driving source	U7 EN is driven by the power switch through diode D6 and R14 (passive circuit).
RFI_N_A, RFI_P_A have no driving source	SX1262 RF inputs are fed from the antenna through the matching network.
IO/Input Pin + Unspecified Port	Port I/O type not specified; connections through A_top verified.
Off-grid Net Label / Port	VCC_3.3V_A, CAN_TERM_A and the SD_CARD sheet port are electrically connected, only not on the 100 mil grid.

Board B

Message	Explanation
GND contains IO Pin and Power Pin	Address/configuration pins tied to GND on purpose: BME280 SDO (address 0x76), LSM6DSOX SA0/SDx/SCx (address 0x6A, Aux unused).
FLASH_SCK has no driving source	The clock goes through series R10 22 Ω; Altium does not see a driver through a resistor.
NetR15_1, NetR16_2, NetR17_2 have no driving source	Configuration inputs set by resistors.
PWR_EN has no driving source	U6/U11 EN is controlled by switch SW3 through R38.
RFI_N, RFI_P have no driving source	SX1262 RF inputs through the matching network.
CAN_RX, RF_MISO: IO Pin + Output Pin	Transceiver/radio output to MCU input. Normal.
IO/Input/Output Pin + Unspecified Port	Port I/O type not specified; connections through B_Top verified.
Off-grid Net Label / Port	ADC_LOOP, BAT_SENSE, RF_SW_CTRL, SW_RFI, SW_RFO, VCC_3.3V, UART_RX, UART_TX are electrically connected, only not on the grid.

Real errors found and fixed during the review: U4 (STM32G474) VDD/VSS pins were swapped in the symbol (MCU without power); U18 LSM6DSOX pins were floating; I2C and SD signals were disconnected from the MCU (ports without net labels); STATUS_LED was not connected to PC5; board A NRST was not connected to the button; B SDA1 moved to PB7 (pin 60), SCL1 to PA15; board A PWR_HOLD_A / PWR_SENSE_A / SD_PWR_EN_A ports reconnected.

LSE crystal 32.768 kHz (A Y1, B Y2)

The new ABS07-32.768KHZ-7-T crystal (CL = 7 pF, ESR ≤ 70 kΩ) works with the existing 10 pF capacitors (A C2/C3, B C21/C22). The previous 830014219B (CL 12.5 pF) had too little gain margin for the STM32F105 oscillator (AN2867).

$$C_L = \frac{C_1 C_2}{C_1 + C_2} + C_{stray} = \frac{10 \cdot 10}{20} + 2 = 7\ \text{pF}$$

$$g_{m,crit} = 4 \cdot ESR \cdot (2\pi f)^2 \cdot (C_0 + C_L)^2 = 4 \cdot 70\,\text{k}\Omega \cdot (2\pi \cdot 32768)^2 \cdot (8.5\ \text{pF})^2 \approx 0.86\ \mu\text{A/V}$$

MCU	Oscillator g_m	Margin / condition	Result
A: STM32F105	5 µA/V	5 / 0.86 ≈ 5.8 (≥ 5 required)	OK
B: STM32G474	gm_crit_max 1.7 µA/V (LSEDRV = MEDIUM-HIGH)	0.86 < 1.7	OK; firmware: RCC_LSEDRIVE_MEDIUMHIGH
Previous 830014219B (CL 12.5 pF)	—	gm_crit ≈ 2.3 µA/V → margin 2.1 < 5	Not OK (hence replaced)

Assumptions: C_stray ≈ 2 pF (pin and trace capacitance), C0 ≈ 1.5 pF. With MEDIUM-LOW drive (0.75 µA/V) the board B crystal might not start.

Converter losses and heating (TPS63021, TPS61023)

All converters heat up by less than 10 °C above ambient, and the TPS61023 output ripple is ≈ 3 mV. Loads are taken from the Power budget section.

$$P_{loss} = P_{out}\left(\frac{1}{\eta} - 1\right), \qquad \Delta T_J = P_{loss} \cdot R_{\theta JA}$$

Converter	P_out	η (assumed)	P_loss	RθJA	ΔT_J
A U7 TPS63021 (3.3 V, 390 mA)	1.29 W	88–90 %	0.14–0.18 W	41.8 °C/W	6–8 °C
B U11 TPS63021 (3.3 V, 426 mA)	1.41 W	88–90 %	0.16–0.19 W	41.8 °C/W	7–8 °C
B U6 TPS61023 (5 V, 122 mA fan)	0.61 W	≈ 90 %	0.07 W	142.7 °C/W	≈ 10 °C

η is read from the datasheet efficiency curves (V_IN 3.0–3.6 V); the exact η will be measured on the prototype. TPS61023 battery current at 3.0 V ≈ 0.61 / 0.9 / 3.0 ≈ 0.23 A.

C40/C41 ripple (B, TPS61023 output). 2 × 10 µF 25 V 0805 at 5 V give ≈ 15 µF effective capacitance (DC bias); TI requires ≥ 4 µF.

$$\Delta V = \frac{I_{out}\,(1 - V_{in}/V_{out})}{f_{sw}\,C_{eff}} = \frac{0.122 \cdot (1 - 3.0/5.0)}{1\ \text{MHz} \cdot 15\ \mu\text{F}} \approx 3.3\ \text{mV}$$

TPS63021 output: 4 × 22 µF 10 V 0805 (≈ 48 µF effective at 3.3 V); inductor 2.2 µH MLP2016H2R2MT0S1: 1.2 A ≥ 0.72 A peak current (SLVS916, Table 1 and Equation 2).

TVS clamping voltages vs. pin limits

The TVS working voltage is above the normal line voltage everywhere, but the clamping voltage at full pulse current exceeds the absolute maximum of the protected pins. This is normal: the TVS diverts most of the pulse energy and the remainder is absorbed by the ICs' internal ESD diodes.

TVS (location)	V_RWM	V_BR	V_CL (8/20 µs pulse)	Protected pin, abs. max	Conclusion
SMAJ5.0CA — A D7, B D4 (VBUS)	5.0 V	6.4–7.0 V	9.2 V @ 43.5 A	MCP73831 VDD 7.0 V	For small pulses the clamp ≈ V_BR ≤ 7 V — at the limit; does not protect against a sustained > 7 V (faulty charger)
SMAJ5.0CA — B D3 (VBAT)	5.0 V	6.4–7.0 V	9.2 V @ 43.5 A	TPS63021 VIN 7 V	OK; leakage at 5 V up to 1.6 mA (bidirectional — 2 × 800 µA), to be measured at 4.2 V
USBLC6-2SC6 / -4SC6 — USB, UART, SWD, UEXT	5 V	≥ 6 V	17 V @ 5 A	STM32 FT pin V_DD + 4 V ≈ 7.3 V	Standard ESD solution; short trace to GND is essential
PESD1CAN — A D2, B D7 (CANH/CANL)	24 V	≥ 25.4 V	35 V @ 1 A, 50 V @ 3 A	SN65HVD230 −4…16 V (transients ±25 V through 100 Ω)	Kept: standard CAN TVS; 12 V alternatives are unidirectional and unsuitable for CAN

Open: B D3 leakage at 4.2 V. At 1 mA a 2 Ah battery would discharge in ≈ 3 months; target < 50 µA, otherwise replace with a lower-leakage TVS.

USB-C current

Both boards draw up to 0.50 A from USB-C: VBUS only supplies the MCP73831 charger. With R_PROG = 2.2 kΩ this fits within the USB 2.0 port limit (500 mA); with any USB-C charger (≥ 1.5 A) it is fine.

- CC1/CC2: R_d = 5.1 kΩ (B R20/R21, RK73H1ETTP5101F) — the board is a sink, the source enables VBUS.
- Allowed current by source: Default USB 2.0 — 500 mA, USB 3.x — 900 mA, Type-C Current — 1.5 A or 3.0 A.
- VBUS load: MCP73831 (A U8, B U12), R_PROG = 2.2 kΩ (A R22, B R17).

$$I_{REG} = \frac{1000\ \mathrm{V}}{R_{PROG}} = \frac{1000}{2200} \approx 0.455\ \mathrm{A}\ (\pm 10\ \% \Rightarrow 0.41\text{–}0.50\ \mathrm{A})$$

Decision (2026-11-09): R_PROG changed from 2 kΩ to 2.2 kΩ so that the current with +10 % tolerance does not exceed 500 mA.

Charge current and heating (MCP73831, BQ24210)

The MCP73831 in SOT-23-5 cannot sustain 0.46 A for long: its thermal regulation will reduce the current. The BQ24210 charges from the solar module at ≈ 0.39 A and heats up by only ≈ 50 °C.

$$P = (V_{IN} - V_{BAT}) \cdot I_{CHG}, \qquad \Delta T_J = P \cdot R_{\theta JA}$$

Charger	Current	P (V_BAT 3.0 V / 3.7 V)	RθJA	ΔT_J	Conclusion
MCP73831 (A U8, B U12), USB 5.0 V, R_PROG 2.2 kΩ	0.455 A	0.91 W / 0.59 W	230 °C/W (≈ 130 with large Cu area)	135–210 °C (77–118 °C with Cu)	Thermal regulation reduces the current; PCB: large Cu area under the IC
BQ24210 (A U6, B U10), solar 5.0 V, R_ISET 1 kΩ	0.39 A (0.37–0.41)	0.78 W / 0.51 W	60.7 °C/W	31–47 °C	OK (T_J ≈ 87 °C at 40 °C ambient)

$$I_{CHG,BQ} = \frac{K_{ISET}}{R_{ISET}} = \frac{390\ \mathrm{A\Omega}}{1\ \mathrm{k\Omega}} = 0.39\ \mathrm{A}$$

MCP73831: allowed dissipation up to 150 °C at 25 °C ambient = (150 − 25) / 230 ≈ 0.54 W, i.e. ≈ 0.27 A at V_BAT 3.0 V and ≈ 0.42 A at 3.7 V. No hazard, only longer charging.

BQ24210 VDPM (A R25, B R22: 1 kΩ → 10 kΩ, 2026-11-09). V_DPM = 3.5 V + 0.15 V/kΩ × R_VDPM. With 1 kΩ this gave 3.65 V — below the 4.2 V charge voltage, so the panel maximum power point was not tracked. For a 5 V-class module (V_mp ≈ 5 V): R_VDPM = (5.0 − 3.5) / 0.15 = 10 kΩ → V_DPM = 5.0 V.

Solar module. The CL-SM3P (3 W) listed earlier has V_mp 17.6 V / V_oc 21.8 V (or 8.8 V / 10.9 V) — not suitable for the BQ24210 (OVP 7 V, VBUS abs. max 20 V). Selected: WSL Solar 5V 1W (WSL-C024) — V_mp 5 V, I_mp 200 mA, V_oc 6 V, I_sc 220 mA, 90 × 90 mm. V_oc 6 V < OVP 7 V (TI recommends V_oc < 7 V) and < SMAJ5.0CA V_BR 6.4 V; V_DPM 5.0 V = V_mp. Two modules in parallel would give ≈ 400 mA ≈ I_SET 390 mA. In the cold (−20 °C) V_oc rises by ≈ 15 % to ≈ 6.9 V, so the solar input TVS (A D9, B D2) may be changed to a unidirectional SMAJ6.5A (V_BR 7.22 V, V_C 11.2 V < 20 V).

AO3401A voltage drop (A Q3, B Q1)

The reverse-polarity P-MOSFET (drain – VBAT, source – VBAT_PROT, gate through 100 kΩ to GND) drops up to 64 mV and heats up by less than 6 °C. V_GS = −V_BAT, so for the worst case (V_BAT 3.0 V) R_DS(on) max at V_GS = −2.5 V is used.

$$V_{DS} = I_{BAT} \cdot R_{DS(on)}, \qquad P = I_{BAT}^2 \cdot R_{DS(on)}$$

Board	I_BAT	R_DS(on) max	V_DS	P	ΔT (125 °C/W)
A Q3	0.47 A	85 mΩ	40 mV	19 mW	2 °C
B Q1	0.75 A	85 mΩ	64 mV	48 mW	6 °C

At least 2.94 V remains at the TPS63021 input — far above its 1.8 V minimum. At V_BAT 4.2 V, R_DS(on) ≤ 60 mΩ, so the drop is even smaller.

SX1262 supply capacitance for the TX burst

The transmitter turn-on step (118 mA, +22 dBm) causes a ≈ 50 mV (1.5 %) dip on the 3.3 V rail — the SX1262 works down to 1.8 V, so the margin is large. Until the TPS63021 responds, the current is supplied by the output and local capacitors.

$$\Delta V = \frac{I_{TX} \cdot \Delta t}{C} = \frac{0.118\ \text{A} \cdot 20\ \mu\text{s}}{\approx 50\ \mu\text{F}} \approx 47\ \text{mV}$$

- C ≈ 50 µF: TPS63021 output 4 × 22 µF (≈ 48 µF effective) + C2 4.7 µF at SX1262 VBAT/VBAT_IO (B U1) + 100 nF.
- Δt ≈ 20 µs — assumed TPS63021 response time (2.4 MHz, PWM mode).
- Firmware: SetTxParams RampTime ≥ 40 µs (e.g. 200 µs) — the current rises more slowly, the converter keeps up and the dip is even smaller.

C8 and C17 filters (board B)

C8 smooths the battery voltage measurement (f_c ≈ 6.8 Hz), C17 smooths the DAC output (f_c ≈ 1.6 kHz). Both are 100 nF.

$$f_c = \frac{1}{2\pi RC}, \qquad \tau = RC$$

Filter	R	C	τ	f_c	Purpose
C8 — BAT_SENSE divider R2/R5 (470 kΩ + 470 kΩ), R4 1 kΩ to ADC	R_th = 470 k ‖ 470 k = 235 kΩ	100 nF	23.5 ms	6.8 Hz	Battery voltage at ADC input
C17 — DAC_OUT → R6 1 kΩ → DAC_FILT → R7 10 kΩ → ADC_LOOP	1 kΩ	100 nF	0.1 ms	1.6 kHz	DAC smoothing, loop-back ADC check

Additional C8 checks:

- Divider 1/2: V_BAT 4.2 V → 2.1 V < V_REF 3.3 V.
- Divider current: 4.2 V / 940 kΩ ≈ 4.5 µA — negligible battery load.
- The ADC sampling capacitor (≈ 5 pF) takes only 5 pF / 100 nF = 0.005 % of the voltage — no error despite the high R_th.
- After power-up, measure no earlier than 5τ ≈ 120 ms.

VDDA filter (board B, C15)

VDDA (U4 pin 29) is fed through ferrite FB1 (BLM21BD272, 2.7 kΩ @ 100 MHz) and decoupled by C15 100 nF ‖ C16 1 µF.

Status of DR1 calculations

- ☑ Power budget for both boards (display and G474 — TBR)
- ☑ Data flow: sensors vs CAN vs radio (group 4)
- ☑ Board B buffer memory capacity
- ☑ Power-loss protection (divider, thresholds, EN levels)
- ☑ RF matching: Zopt, L1/C1, H2 trap, π filter, band
- ☑ I2C pull-up limits (SLVA689)
- ☑ SPI timing budget for the SD card
- ☑ Lecturer’s list (2026-11-08): LSE, converter heating, TVS, USB-C, charging, AO3401A, SX1262 TX, C8/C17
- ☑ Decisions (2026-11-09): R_VDPM 10 kΩ, R_PROG 2.2 kΩ, PESD1CAN kept, 5 V solar module
- ☑ Final ERC (2026-11-09): A and B — 0 errors
- ☐ TBR values: display current (measure), G474 current (CubeMX), SD output delay, F105 t_su(MI), Fox crystal ESR/C0
- ☐ Open: B D3 leakage at 4.2 V (measure on prototype)